

2026_Baes_JRF_HRP_AI_A_web_applicaition_for_Hypertension_Risk_Prediction_with_Calibrated_Explainable...
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Abstract

Hypertension remains a major public health concern in the Philippines, contributing significantly to cardiovascular morbidity and mortality. This study develops HRP AI, a web-based decision-support application for predicting clinical hypertension risk ($\geq 140/90$ mmHg) among Filipino adults using calibrated and explainable machine learning.

Using the DOST-FNRI 2015 National Nutrition Survey (NNS), a final dataset of 151,189 adults was constructed, with 22,988 hypertensive cases corresponding to a 15.20% prevalence rate. Eight machine learning algorithms were evaluated across multiple imbalance-handling configurations through a Two-Stage Randomized Search. XGBoost with algorithmic Class Weights was selected as the final predictive engine because it provided the strongest clinical compromise between sensitivity, discrimination, and calibration compatibility. At the standard decision threshold, the model achieved

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